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Telecommunication Systems I
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Exercise 2
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Hours required: 3.5 hours

PART A

A. In the first part of the exercise, which is mainly experimental, we will study the spectral content of baseband PAM waveforms.

A.1 Construct an SRRC pulse $\phi(t)$ with parameters $T = 10^{-3}$ sec, $\text{over} = 10$, $T_s = \frac{T}{\text{over}}$, $A = 4$, and $a = 0.5$.

(10) Using the functions `fftshift` and `fft`, compute the magnitude of the Fourier transform of $\phi(t)$, $|\Phi(F)|$, at N_f equally spaced points in the interval $[-\frac{F_s}{2}, \frac{F_s}{2}]$.¹ Plot the energy spectral density $|\Phi(F)|^2$ versus the appropriate frequency axis using semilogy.

Matlab code:

```
% parameters
T      = 10^-3;
over   = 10;
Ts     = T/over;
Fs     = 1 / Ts;
A      = 4;
a      = 0.5;
Nf     = 2048;

%% A.1
[phi, t_phi] = srrc_pulse(T, over, A, a);
F = (-Nf/2:Nf/2-1)*(Fs/Nf);
Phi = Ts*fftshift(fft(phi, Nf));
ESD_phi = abs(Phi).^2;

figure;
semilogy(F, ESD_phi); grid on;
xlabel('f (Hz)'); ylabel('|Phi(F)|^2');
title('ESD of SRRC pulse');
```

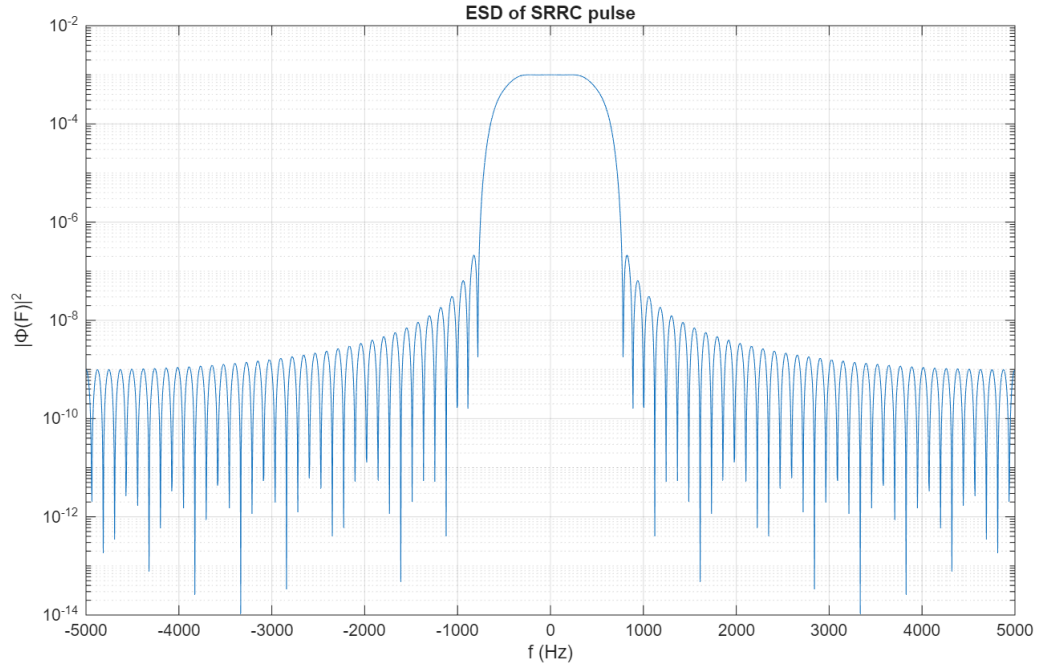


Figure 1. Energy spectral density $|\Phi(F)|^2$ of the SRRC pulse.

A.2 Generate a sequence of $N = 100$ independent and equiprobable bits $\{b_0, \dots, b_{N-1}\}$.

Using the mapping

$$0 \longrightarrow +1,$$

$$1 \longrightarrow -1,$$

map the bits to symbols X_n , for $n = 0, \dots, N - 1$.

Construct the waveform

$$X(t) = \sum_{n=0}^{N-1} X_n \phi(t - nT).$$

Assuming an infinite number of symbols, we proved that the power spectral density of $X(t)$ is

$$S_X(F) = \frac{\sigma_X^2}{T} |\Phi(F)|^2.$$

MATLAB code:

```
%% A.2
N = 100; % Number of bits
bits = randi([0 1], 1, N); % Generate N independent bits

% Mapping
Xn = 1 - 2*bits;

[phi, t_phi] = srrc_pulse(T, over, A, a);

Lx = (N-1)*over + length(phi); % Length of the final waveform in samples

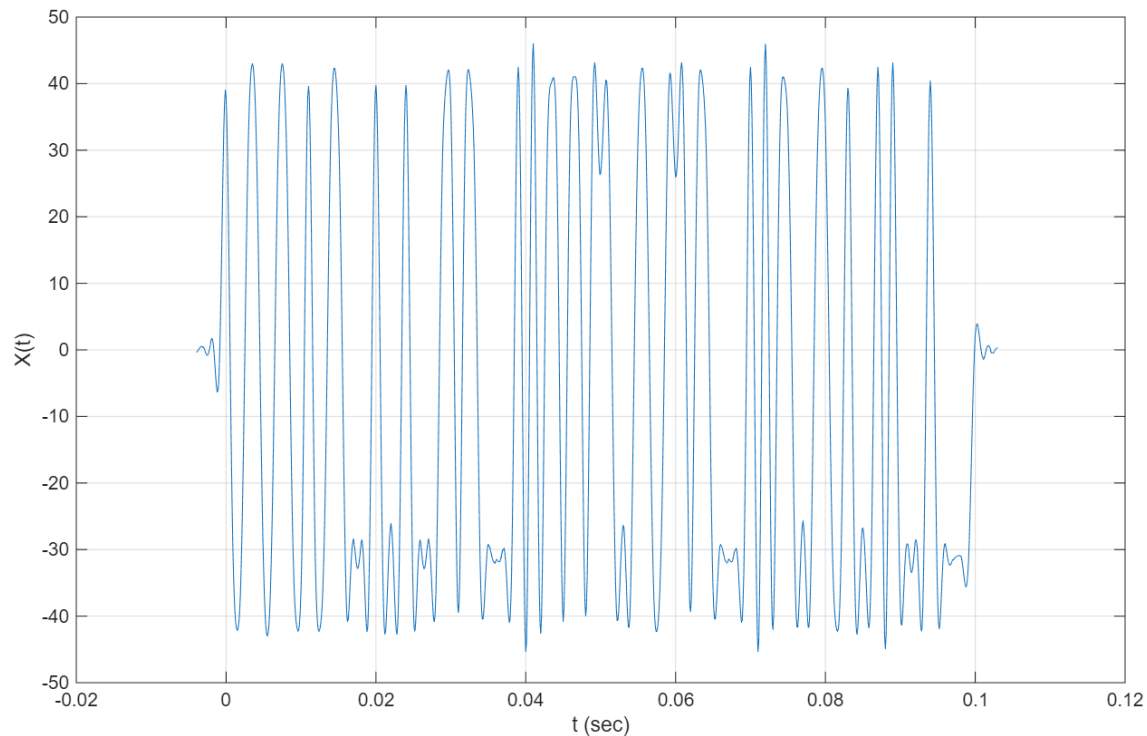
% Initialize waveform
x = zeros(1, Lx);

% X(t) = sum Xn * phi(t - nT)
for n = 0:N-1
    start_index = n*over + 1;
    end_index = start_index + length(phi) - 1;

    x(start_index:end_index) = x(start_index:end_index) + Xn(n+1)*phi;
end

t_x = (0:Lx-1)*Ts + t_phi(1); % Time axis for X(t)

figure;
plot(t_x, x); grid on;
xlabel('t (sec)'); ylabel('X(t)');
```



A.3 (10) Using the functions `fft` and `fftshift`, compute the periodogram of one realization of $X(t)$:

$$P_X(F) = \frac{|\mathcal{F}[X(t)]|^2}{T_{\text{total}}},$$

where T_{total} is the **total time duration** of $X(t)$ in sec. Plot $P_X(F)$ using both plot and semilogy.

Repeat for several different realizations of the bit sequence $\{b_0, \dots, b_{N-1}\}$, so as to get a good sense of what the periodograms of realizations of $X(t)$ look like.

MATLAB code:

```
% A.3
Nf = 2048;
Fs = 1/Ts;
Ttotal = length(x) * Ts;    % Total duration

% Frequency axis from -Fs/2 to Fs/2
F = (-Nf/2:Nf/2-1) * Fs/Nf;

% Fourier transform of X(t)
X_F = Ts * fftshift(fft(x, Nf));

% Periodogram
PX = abs(X_F).^2 / Ttotal;

% Plot
figure;
plot(F, PX); grid on;
xlabel('f (Hz)'); ylabel('P_X(F)');
title('Periodogram of one realization of X(t) (linear scale)');

% semilogy
figure;
semilogy(F, PX); grid on;
xlabel('f (Hz)'); ylabel('P_X(F)');
title('Periodogram of one realization of X(t) (semilogy scale)');
```

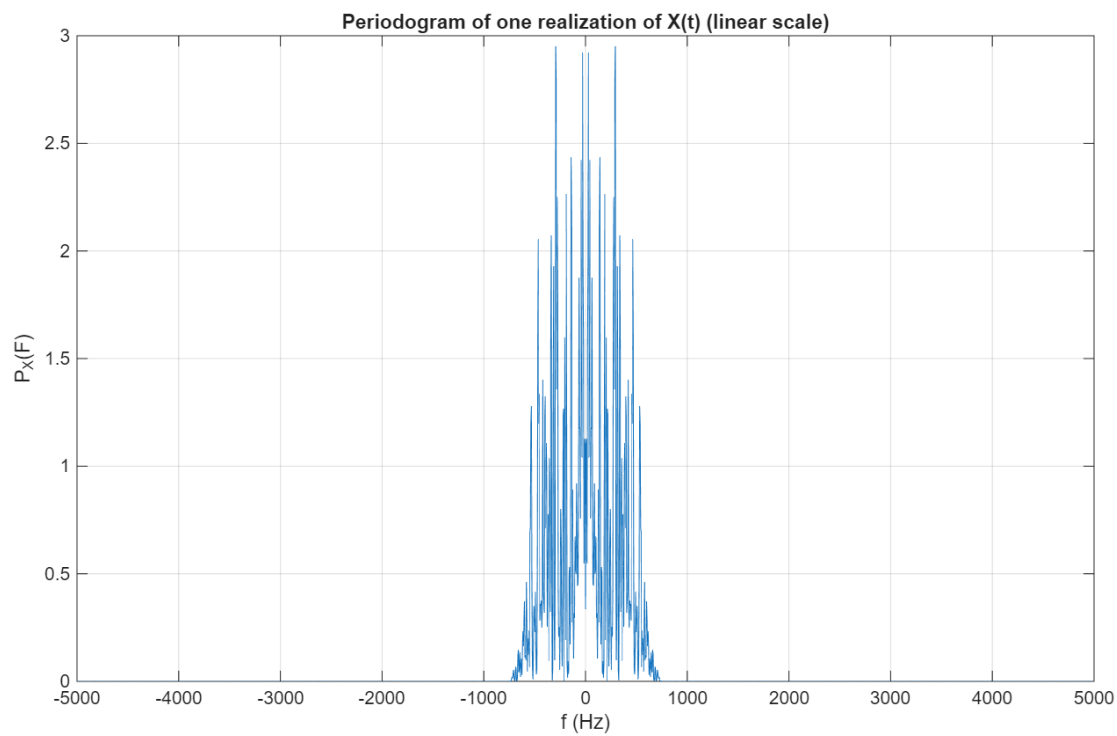


Figure 3. Periodogram of one 2-PAM realization, linear scale.

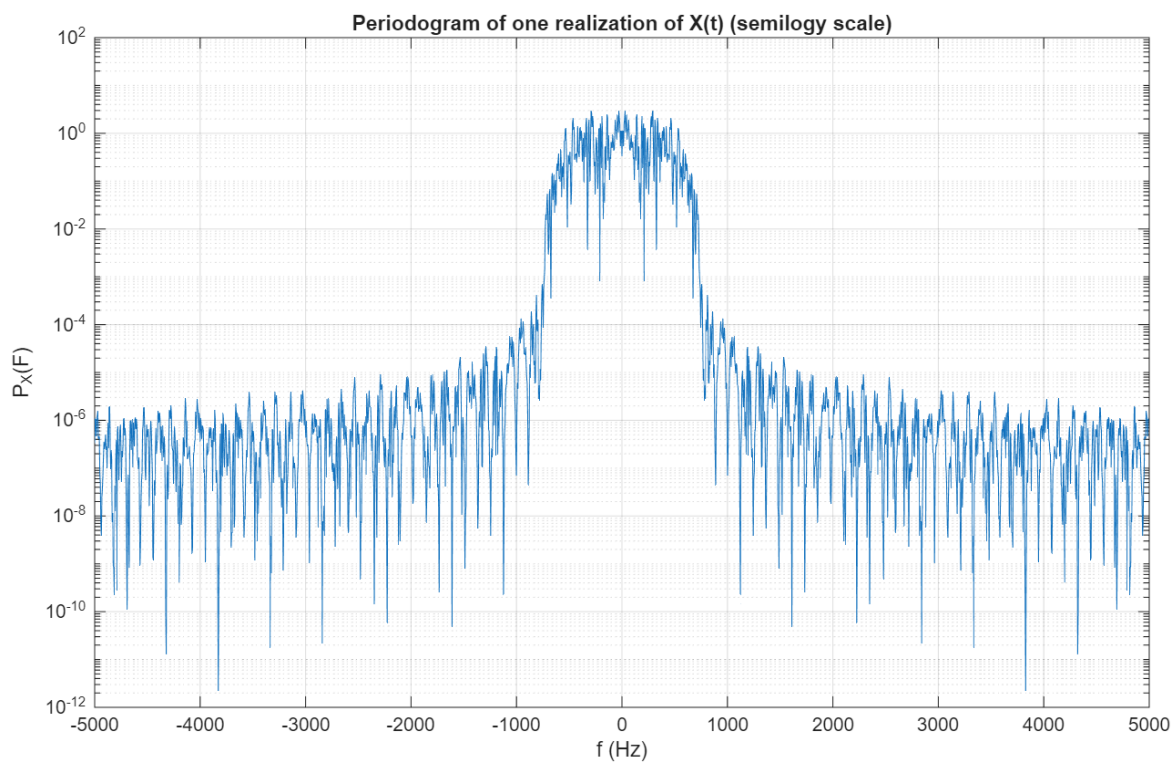


Figure 4. Periodogram of one 2-PAM realization, semilogarithmic scale.

(10) Estimate the power spectral density by computing numerical averages over K (indicatively, $K = 500$) periodogram realizations. On the same semilogy plot, show both the estimate and the theoretical² power spectral density.

MATLAB code:

```
%% A.3 - b

K = 500;
PX_sum = zeros(1, Nf);

for k = 1:K

    bits = randi([0 1], 1, N);
    Xn = 1 - 2*bits;

    x = zeros(1, Lx);

    for n = 0:N-1
        start_index = n*over + 1;
        end_index = start_index + length(phi) - 1;

        x(start_index:end_index) = x(start_index:end_index) + Xn(n+1)*phi;
    end

    X_F = Ts * fftshift(fft(x, Nf));
    PX = abs(X_F).^2 / Ttotal;

    PX_sum = PX_sum + PX;
end

PX_est = PX_sum / K;

sigma2_X = 1;
Phi = Ts * fftshift(fft(phi, Nf));
SX_theory = (sigma2_X / T) * abs(Phi).^2;

figure;
semilogy(F, PX_est);
hold on;
semilogy(F, SX_theory, '--');
grid on;

xlabel('f (Hz)');
ylabel('PSD');
title('Estimated and theoretical PSD');
legend('Estimated PSD, K = 500', 'Theoretical PSD');
```

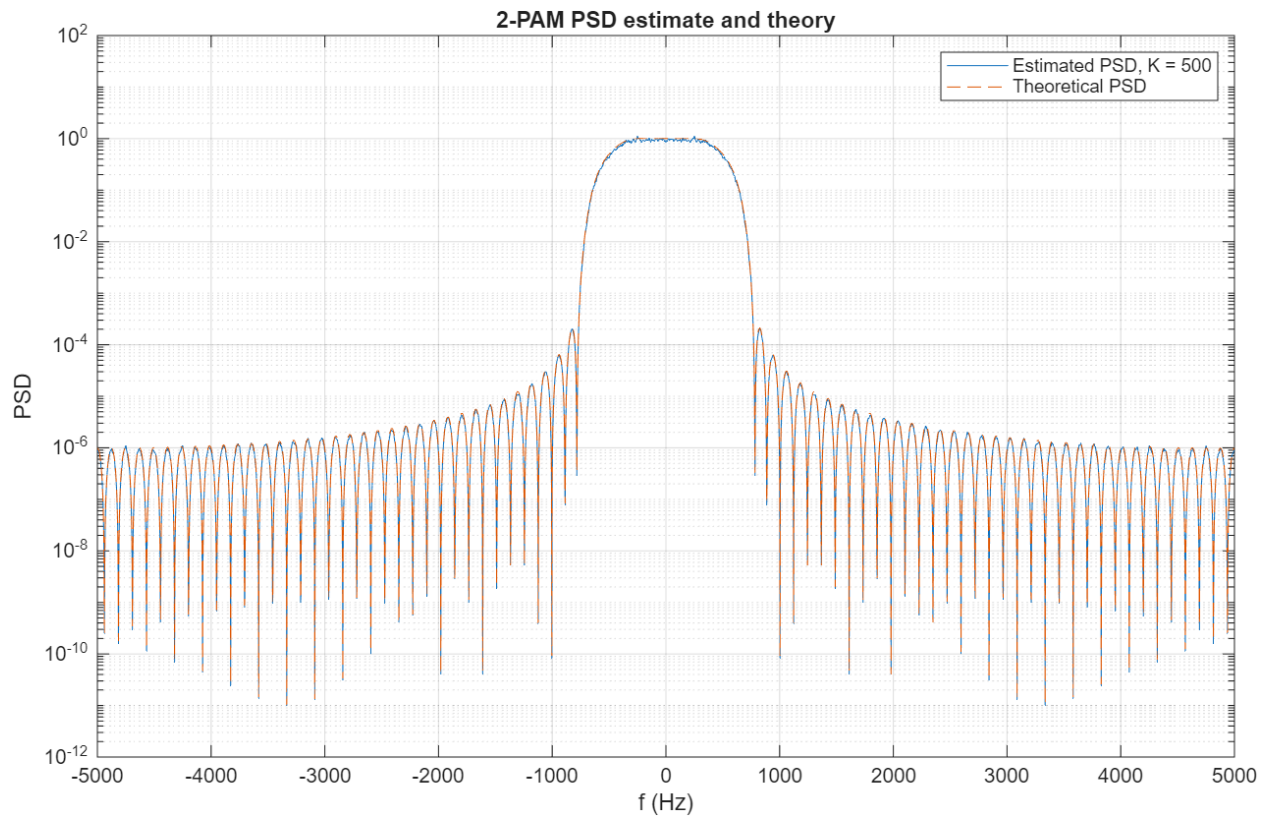


Figure 5. Average periodogram estimate for 2-PAM compared with the theoretical PSD.

(10) As you increase K and N , the approximation should improve. Does this happen in your experiments? If yes, can you explain why?

Yes, the approximation improves as K and N increase.

Increasing K means that more independent periodograms are averaged, so the random fluctuations of each individual periodogram are reduced. Increasing N increases the duration of the observed waveform, which improves the frequency resolution and makes the periodogram a better approximation of the true power spectral density. Therefore, for larger K and N , the estimated PSD becomes smoother and closer to the theoretical PSD.

A.4 Using the mapping

$$00 \longrightarrow +3$$

$$01 \longrightarrow +1$$

$$11 \longrightarrow -1$$

$$10 \longrightarrow -3$$

construct the 4-PAM symbol sequence X_n , for $n = 0, \dots, \frac{N}{2} - 1$. Note that, *if the bits are equiprobable, then the symbols X_n are also equiprobable!*

Construct the waveform

$$X(t) = \sum_{n=0}^{\frac{N}{2}-1} X_n \phi(t - nT)$$

using the same symbol period T as in question A.2.

(10) Compute the periodogram and estimate the power spectral density by averaging periodogram realizations of $X(t)$. Plot the experimental and the theoretical power spectral density on the same semilogy. What do you observe?

MATLAB code:

```
% A.4 - 4-PAM waveform and PSD estimation
```

```
% Generate N independent equiprobable bits
```

```
bits = randi([0 1], 1, N);
```

```
Nsym4 = N/2; % Number of 4-PAM symbols
```

```
% Initialize 4-PAM symbol sequence
```

```
Xn4 = zeros(1, Nsym4);
```

```
% Mapping
```

```
for n = 1:Nsym4
```

```
    b1 = bits(2*n - 1);
```

```
    b2 = bits(2*n);
```

```
    if b1 == 0 && b2 == 0
```

```
        Xn4(n) = 3;
```

```
    elseif b1 == 0 && b2 == 1
```

```
        Xn4(n) = 1;
```

```
    elseif b1 == 1 && b2 == 1
```

```
        Xn4(n) = -1;
```

```
    elseif b1 == 1 && b2 == 0
```

```
        Xn4(n) = -3;
```

```
    end
```

```
end
```



```

% Construct 4-PAM waveform
Lx4 = (Nsym4 - 1)*over + length(phi);
x4 = zeros(1, Lx4);

for n = 0:Nsym4-1
    start_index = n*over + 1;
    end_index = start_index + length(phi) - 1;

    x4(start_index:end_index) = x4(start_index:end_index) + Xn4(n+1)*phi;
end

% Time axis for 4-PAM waveform
t_x4 = (0:Lx4-1)*Ts + t_phi(1);

Ttotal4 = length(x4) * Ts;

% Fourier transform of x4
X4_F = Ts * fftshift(fft(x4, Nf));

% Periodogram
PX4 = abs(X4_F).^2 / Ttotal4;

%% PSD estimation by averaging K periodograms

K = 500;
PX4_sum = zeros(1, Nf);

for k = 1:K

    % Generate new random bits
    bits = randi([0 1], 1, N);

    % Convert bits to 4-PAM symbols
    Xn4 = zeros(1, Nsym4);

    for n = 1:Nsym4

        b1 = bits(2*n - 1);
        b2 = bits(2*n);

        if b1 == 0 && b2 == 0
            Xn4(n) = 3;
        elseif b1 == 0 && b2 == 1
            Xn4(n) = 1;
        elseif b1 == 1 && b2 == 1
            Xn4(n) = -1;
        elseif b1 == 1 && b2 == 0
            Xn4(n) = -3;
        end
    end

    % Construct 4-PAM waveform
    x4 = zeros(1, Lx4);

    for n = 0:Nsym4-1

```

```

        start_index = n*over + 1;
        end_index = start_index + length(phi) - 1;

        x4(start_index:end_index) = x4(start_index:end_index) + Xn4(n+1)*phi;
    end

    % Fourier transform
    X4_F = Ts * fftshift(fft(x4, Nf));

    % Periodogram
    PX4 = abs(X4_F).^2 / Ttotal4;

    % Add to sum
    PX4_sum = PX4_sum + PX4;
end

% Estimated PSD
PX4_est = PX4_sum / K;

% Theoretical PSD for 4-PAM
sigma2_X4 = 5;
Phi = Ts * fftshift(fft(phi, Nf));
SX4_theory = (sigma2_X4 / T) * abs(Phi).^2;

% plot
figure;
semilogy(F, PX4_est); hold on;
semilogy(F, SX4_theory, '--');
grid on;

xlabel('f (Hz)'); ylabel('PSD');
title('Estimated and theoretical PSD for 4-PAM');
legend('Estimated PSD, 4-PAM', 'Theoretical PSD, 4-PAM');

```

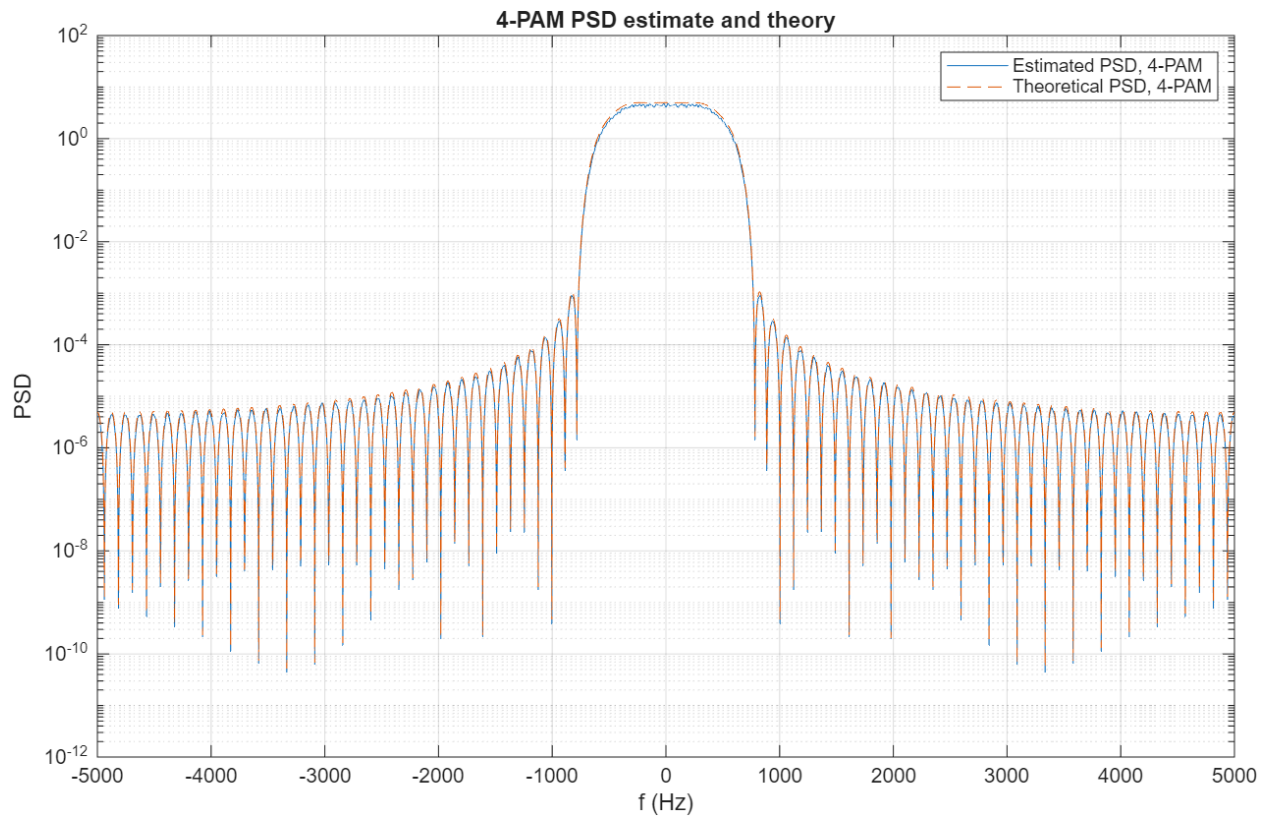


Figure 6. Average periodogram estimate for 4-PAM compared with the theoretical PSD.

The estimated PSD obtained by averaging $K = 500$ periodograms follows the theoretical PSD quite well. The shape of the PSD is determined by the spectrum of the SRRC pulse. Since the symbol period T and the pulse shape $\phi(t)$ are the same as in the 2-PAM case, the bandwidth remains approximately the same. However, the magnitude of the PSD is larger.

(10) How does the power spectral density of this $X(t)$ compare (in terms of bandwidth and peak magnitude) to the power spectral density of the $X(t)$ in step A.2? Can you explain your observations?

The 4-PAM PSD has approximately the same bandwidth as the 2-PAM PSD, because the symbol period T and pulse shape $\phi(t)$ are the same. However, its peak magnitude is larger. This happens because the variance of the 4-PAM symbols is $\sigma_X^2 = \frac{3^2 + 1^2 + (-1)^2 + (-3)^2}{4} = \frac{20}{4} = 5$, while for 2-PAM it is $\sigma_X^2 = \frac{1^2 + (-1)^2}{2} = 1$. Since the PSD is proportional to σ_X^2 , the 4-PAM PSD is about five times larger in magnitude.

A.5 (10) Repeat step A.3 by setting the symbol period to $T' = 2T$ (keep the sampling period T_s the same as in the previous steps; therefore, you must double the parameter over).

(5) What do you observe about the bandwidth of the waveforms in this case compared to that of the waveforms in step A.3? Can you explain the phenomenon?

MATLAB code:

```
%% A.5 2-PAM with Tprime = 2T, same Ts
T5 = 2*T;
over5 = 2*over;
Nf = 4096;

% Frequency axis from -Fs/2 to Fs/2
F = (-Nf/2:Nf/2-1) * Fs/Nf;

% New SRRC pulse for T'
[phi5, t_phi5] = srrc_pulse(T5, over5, A, a);

% Lengths
Lphi5 = length(phi5);
Lx5 = (N-1)*over5 + Lphi5;

% One realization
bits = randi([0 1], 1, N);

% 2-PAM mapping
Xn = 1 - 2*bits;

% Construct waveform
x5 = zeros(1, Lx5);

for n = 0:N-1
    start_index = n*over5 + 1;
    end_index = start_index + Lphi5 - 1;

    x5(start_index:end_index) = x5(start_index:end_index) + Xn(n+1)*phi5;
end

Ttotal5 = length(x5)*Ts;

% Periodogram
X5_F = Ts * fftshift(fft(x5, Nf));
PX5 = abs(X5_F).^2 / Ttotal5;

figure;
semilogy(F, PX5); grid on;
xlabel('f (Hz)'); ylabel('P_X(F)');
title('Periodogram of one realization, T' = 2T');
```

```
%% A.5 - PSD estimation by averaging K periodograms
```

```
K = 500;
PX5_sum = zeros(1, Nf);

for k = 1:K

    % Generate new random bits
    bits = randi([0 1], 1, N);

    % 2-PAM mapping
    Xn = 1 - 2*bits;

    % Construct waveform
    x5 = zeros(1, Lx5);

    for n = 0:N-1
        start_index = n*over5 + 1;
        end_index = start_index + Lphi5 - 1;

        x5(start_index:end_index) = x5(start_index:end_index) + Xn(n+1)*phi5;
    end

    % Fourier transform
    X5_F = Ts * fftshift(fft(x5, Nf));

    % Periodogram
    PX5 = abs(X5_F).^2 / Ttotal5;

    % Add to sum
    PX5_sum = PX5_sum + PX5;
end

% Estimated PSD
PX5_est = PX5_sum / K;

% Theoretical PSD for  $T' = 2T$ 
Phi5 = Ts * fftshift(fft(phi5, Nf));
SX5_theory = (1 / T5) * abs(Phi5).^2;

figure;
semilogy(F, PX5_est);
hold on;
semilogy(F, SX5_theory, '--');
grid on;

xlabel('f (Hz)'); ylabel('PSD');
title('2-PAM PSD for  $T' = 2T$  estimate and theory');
legend('Estimated PSD,  $T' = 2T$ ', 'Theoretical PSD,  $T' = 2T$ ');
```

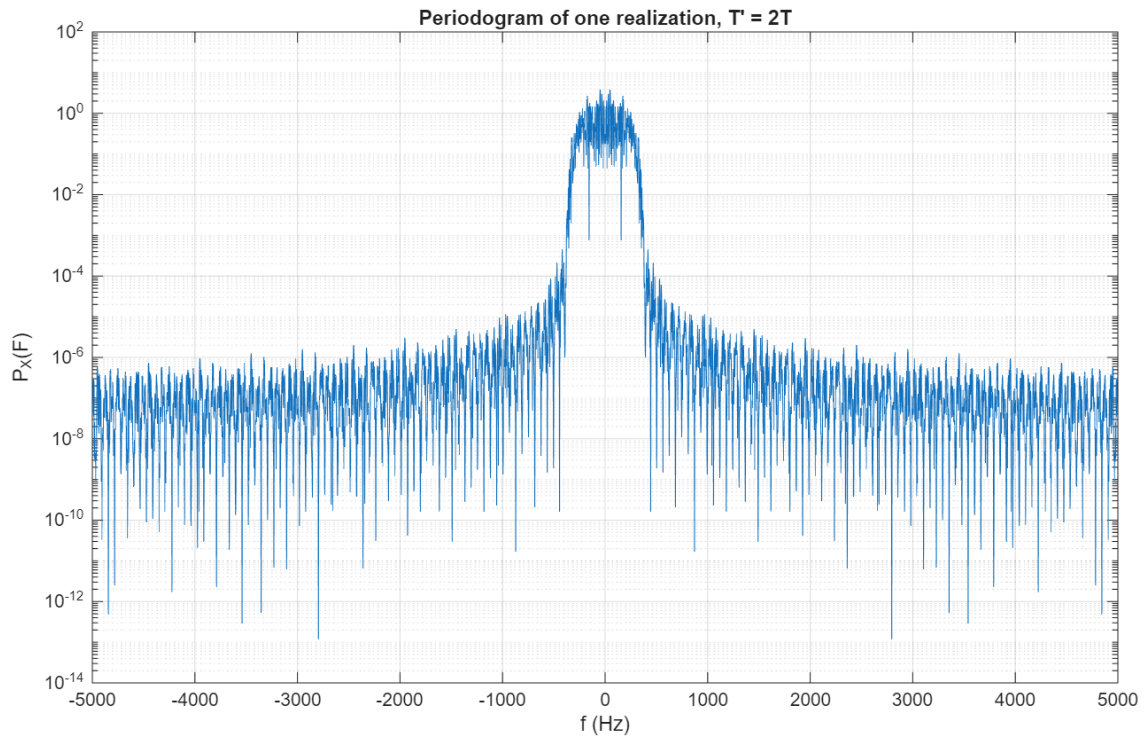


Figure 7. Periodogram of one 2-PAM realization for $T'=2T$, semilogarithmic scale.

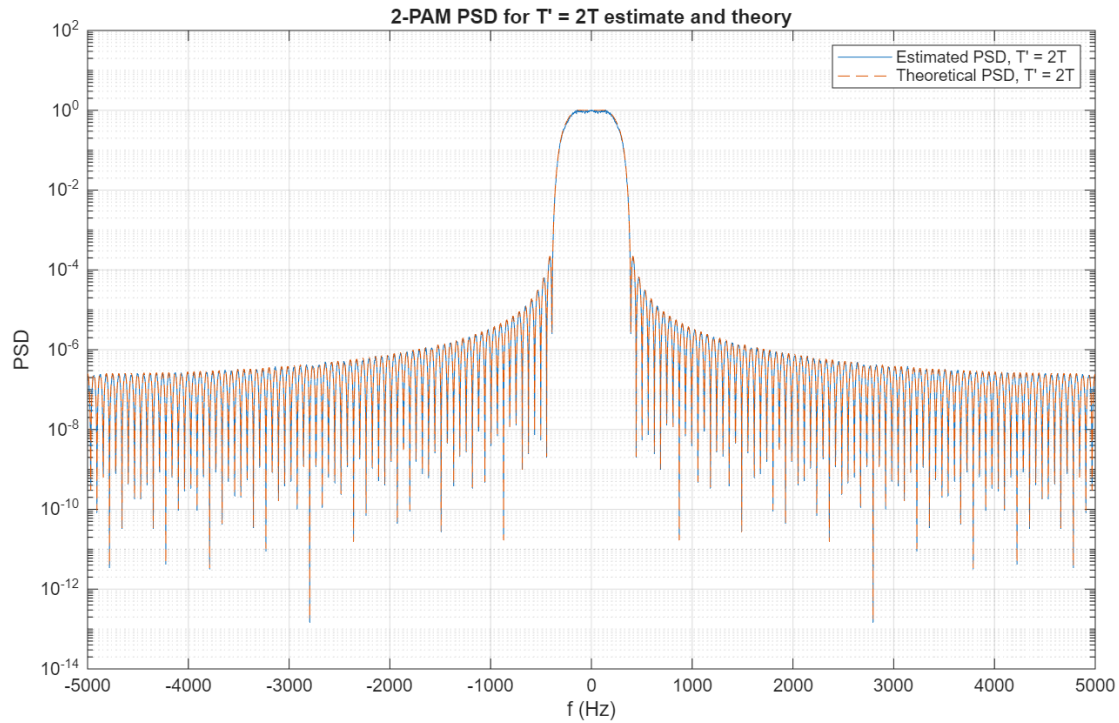


Figure 8. Average periodogram estimate for 2-PAM with $T'=2T$ compared with theory.

The occupied bandwidth is approximately halved when T is doubled. Thus, for $T=1$ ms and $\alpha=0.5$ the one-sided support is about 750 Hz, while for $T'=2$ ms it becomes about 375 Hz.

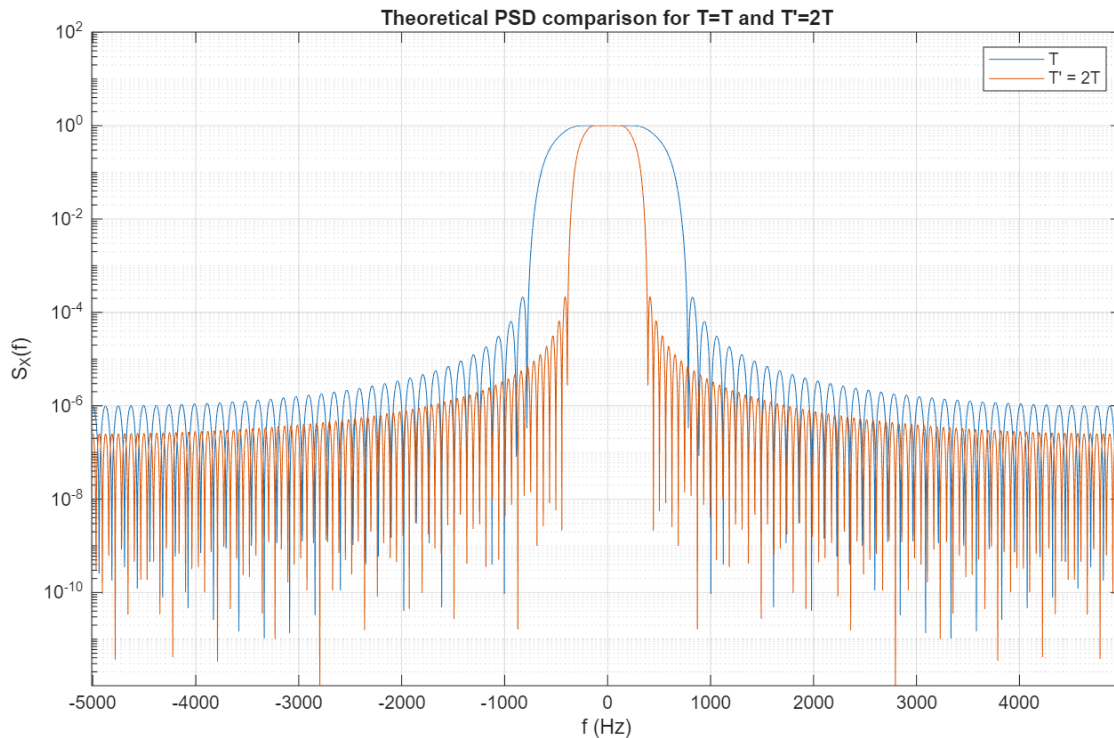


Figure 9. Theoretical PSD comparison for T and $T'=2T$.

Comparing the PSD of A.5 with the PSD of A.3, we observe that the waveform with $T' = 2T$ occupies a smaller bandwidth. In fact, the bandwidth is approximately half of the bandwidth in A.3. This happens because increasing the symbol period decreases the symbol rate, so the transmitted waveform varies more slowly in time and its spectrum becomes narrower.

A.6 (2.5) If you wanted to transmit data as fast as possible while using the same available bandwidth, would you choose 2-PAM or 4-PAM, and why?

(2.5) If the available bandwidth is very expensive, would you choose symbol period T or $T' = 2T$, and why?

I would choose 4-PAM if I wanted to transmit data as fast as possible using the same available bandwidth. This is because 4-PAM transmits 2 bits per symbol, while 2-PAM transmits only 1 bit per symbol. Since the symbol period and bandwidth can remain the same, 4-PAM gives a higher bit rate.

For the second question, I would choose $T' = 2T$ if the available bandwidth is very expensive. Doubling the symbol period reduces the symbol rate, so the signal varies more slowly and occupies a smaller bandwidth. So, $T' = 2T$ is more bandwidth-efficient, although it gives a lower data rate than using T .

PART B

B. In the second part of the exercise, which is mainly theoretical, we will study simple stochastic processes.

Let

$$Y(t) = X \cos(2\pi F_0 t + \Phi), \quad (1)$$

where $X \sim \mathcal{N}(0, 1)$, $\Phi \sim \mathcal{U}[0, 2\pi)$, and X and Φ are independent random variables.

1. (5) Plot, on the same plot, 5 realizations of the process (only for this question, assume $F_0 = 10$ Hz, $T_s = 10^{-2}$ sec, and $t_{\text{axis}} = [0 : T_s : 5/F_0]$).

1.

MATLAB code:

```
% B.1
F0 = 10;
TsB = 10^-2;
t_axis = 0:TsB:5/F0;

figure; hold on; grid on;
for m = 1:5
    X = randn;
    Phi = 2*pi*rand;
    Y = X*cos(2*pi*F0*t_axis + Phi);
    plot(t_axis, Y);
end

xlabel('t (sec)'); ylabel('Y(t)');
title('5 realizations of Y(t)');
```

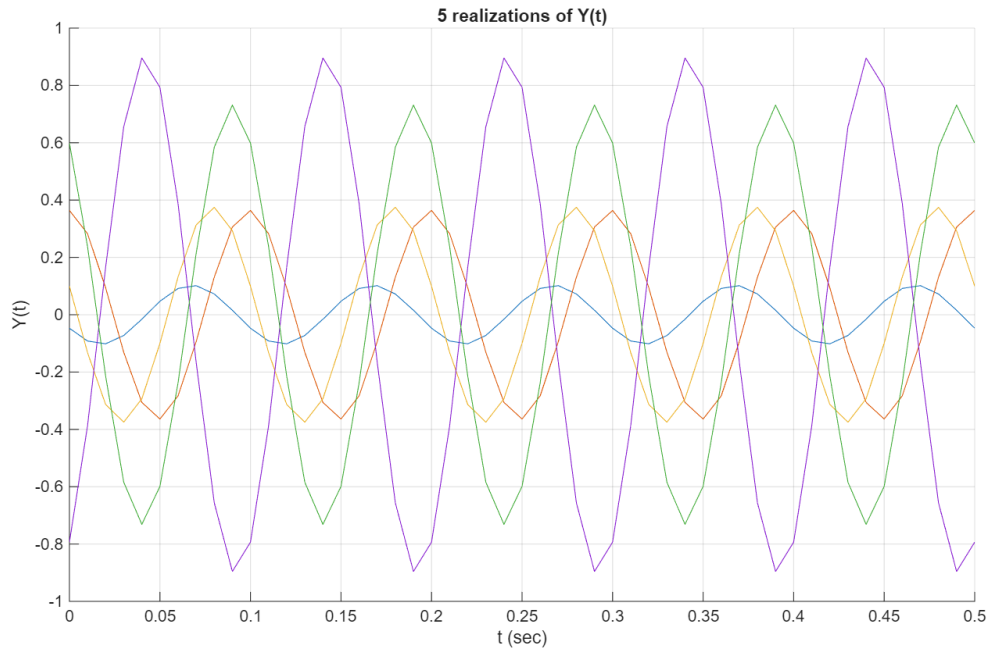


Figure 10. Five realizations of $Y(t)$ for $F_0=10$ Hz, $T_s=10^{-2}$ s, and t from 0 to $5/F_0$.

2. (10) Compute the quantities $\mathcal{E}[Y(t)]$ and $R_{YY}(t + \tau, t) = \mathcal{E}[Y(t + \tau) Y(t)]$. What do you conclude?

2.

The process is:

$$Y(t) = X \cos(2\pi F_0 t + \Phi),$$

where $X \sim N(0,1)$, $\Phi \sim U[0,2\pi)$, and X and Φ are independent.

The mean is:

$$\mathcal{E}[Y(t)] = \mathcal{E}[X \cos(2\pi F_0 t + \Phi)]$$

Since X and Φ are independent,

$$\mathcal{E}[Y(t)] = \mathcal{E}[X] * \mathcal{E}[\cos(2\pi F_0 t + \Phi)]$$

because $\mathcal{E}[X] = 0$, we get

$$\mathcal{E}[Y(t)] = 0$$

The autocorrelation is:

$$R_{YY}(t + \tau, t) = \mathcal{E}[Y(t + \tau) * Y(t)]$$

We have

$$Y(t + \tau) = X \cos(2\pi F_0(t + \tau) + \Phi)$$

and

$$Y(t) = X \cos(2\pi F_0 t + \Phi)$$

Therefore,

$$Y(t + \tau) Y(t) = X^2 \cos(2\pi F_0(t + \tau) + \Phi) \cos(2\pi F_0 t + \Phi)$$

So,

$$R_{YY}(t + \tau, t) = \mathcal{E}[Y(t + \tau) * Y(t)] = \mathcal{E}[X^2 \cos(2\pi F_0(t + \tau) + \Phi) \cos(2\pi F_0 t + \Phi)]$$

Since X and Φ are independent and $\mathcal{E}[X^2] = 1$,

$$R_{YY}(t + \tau, t) = \mathcal{E}[X^2] \mathcal{E}[\cos(2\pi F_0(t + \tau) + \Phi) \cos(2\pi F_0 t + \Phi)]$$

Using

$$\cos A * \cos B = \frac{1}{2} \cos(A - B) + \frac{1}{2} \cos(A + B)$$

with

$$A = 2\pi F_0(t + \tau) + \Phi \text{ and } B = 2\pi F_0 t + \Phi$$

We get

$$R_{YY}(t + \tau, t) = \frac{1}{2} \cos(2\pi F_0 \tau) + \frac{1}{2} \mathcal{E}[\cos(2\pi F_0(2t + \tau) + 2\Phi)]$$

Since Φ is uniform in $[0, 2\pi)$, the second expectation is zero. Thus,

$$R_{YY}(\tau) = \frac{1}{2} \cos(2\pi F_0 \tau)$$

The mean is constant and the autocorrelation depends only on τ , not on t .

Therefore, **$Y(t)$ is wide-sense stationary.**

3. (5) Compute the power spectral density $S_Y(F)$.

3.

Since $Y(t)$ is wide-sense stationary, its power spectral density is the Fourier transform of its autocorrelation:

$$S_Y(F) = \mathcal{F}\{R_{YY}(\tau)\}$$

From B.2

$$R_{YY}(\tau) = \frac{1}{2} \cos(2\pi F_0 \tau)$$

Also,

$$\mathcal{F}\{\cos(2\pi F_0 \tau)\} = \frac{1}{2} [\delta(F - F_0) + \delta(F + F_0)]$$

Therefore,

$$S_Y(F) = \frac{1}{2} \cdot \frac{1}{2} [\delta(F - F_0) + \delta(F + F_0)]$$

So,

$$\mathbf{S_Y(F) = \frac{1}{4} [\delta(F - F_0) + \delta(F + F_0)]}$$